Contents for Summer Internship on Data science with AI using ML

Training Session Plan (100 Hrs.)			Trainer Name : Mr. Himanshu Sardana		
Day	**Module**	**Topic**	**Session Duration (In Hrs.)**	**Training Date**	**Trainer Signature**
Day 1	Python Basics for Data Science	- Introduction to Data Science: What is Data Science, its lifecycle, real- world applications - Introduction to AI and Machine Learning: Types, applications, and their role in Data Science - Python installation, IDEs, Jupyter Notebook walkthrough - Data Types: int, float, str, list, tuple, dict, set - Variables, Identifiers, Naming Conventions - Operators: Arithmetic, Logical, Comparison - Type Casting and User Input	4	2/6/2025	
Day 2	Control Structures and Data Structures	- Conditional Statements: if, else, elif with examples - Loops: for, while, loop control (break, continue, pass) - Data Structures: Lists, Tuples, Sets, Dictionaries with use cases - List Comprehension: Syntax, conditions, nested lists - Practical exercises using control flows and data structures	4	3/6/2025	
Day 3	Functions and File Handling	- Defining and Calling Functions - Arguments and Return Values - Lambda Functions: Syntax and use cases - Built-in functions: map, filter, reduce - Exception Handling: try, except, finally - File Handling: open(), read(), write(), close(), file modes	4	4/6/2025	
Day 4	Python Libraries for Data Science	- NumPy: Creating arrays, indexing, slicing, mathematical operations - Pandas: Series vs DataFrames, importing/exporting data, handling missing data	4	5/6/2025	
Module-1 Assessment Python Programming		**- Hands-on coding tasks and quizzes based on Days 1-4** **- Mini Project: Read a CSV, clean data, and**	**5/6/2025 2:30-4:30 (Group-A) 10:00-12:00 (Group-B)**		
Day 5	Data Cleaning and Handling Missing Data	- Identifying nulls and outliers - Techniques: dropna, fillna, replace - Handling categorical variables (label encoding, one-hot encoding) - Data consistency and integrity checks	4	6/6/2025	

Day 6	Data Manipulation and Aggregation	- Data Selection using loc and iloc - GroupBy operations for aggregation - Merging and Joining DataFrames - Pivot Tables and Cross-tabulations - Reshaping with melt and stack/unstack	4	7/6/2025	
Day 7	Data Visualization Basics	- Matplotlib: Plot types, figure layout, custom styling - Seaborn: Univariate and bivariate analysis - Plot customization: labels, titles, legends, annotations - Use of pairplots, heatmaps for correlation	4	9/6/2025	
Day 8	Exploratory Data Analysis (EDA)	- Step-by-step EDA workflow - Summary statistics and distribution analysis - Correlation matrix and pairwise relationships - Feature engineering: binning, scaling, transformation	4	10/6/2025	
Module-2 Assessment Data Analysis and Visualization		**- EDA mini project: Analyze and visualize a dataset** **- Quiz on Pandas, NumPy, Matplotlib, Seaborn**	**10/6/2025** **2:30-4:30 (Group-A)** **10:00-12:00 (Group-B)**		
Day 9	Descriptive Statistics	- Measures of Central Tendency: mean, median, mode - Measures of Dispersion: range, variance, standard deviation, IQR - Data distributions: symmetric, skewed	4	11/6/2025	
Day 10	Probability and Distributions	- Basic probability rules and formulas - Conditional probability and Bayes Theorem - Random variables and probability distributions - Binomial, Normal, Poisson distributions – use cases	4	12/6/2025	
Day 11	Hypothesis Testing	- Statistical Hypotheses: null vs alternative - Test types: One-sample, two-sample, paired - p-values, significance levels - Tests: t-test, chi-square, ANOVA	4	13/6/2025	
Module-3 Assessment Statistics and Probability		**- MCQs on stats and probability theory** **- Case-based problems requiring hypothesis testing**	**13/6/2025** **2:30-4:30 (Group-A)** **10:00-12:00 (Group-B)**		
Day 12	Introduction to Machine Learning	- ML workflow: data collection, preprocessing, training, evaluation, deployment - Supervised vs Unsupervised learning - Model selection and training process using Scikit-learn - Data preprocessing: scaling, encoding, splitting	4	14/6/2025	
Day 13	Supervised Learning - Regression	- Linear Regression: coefficients, intercept, assumptions - Polynomial regression and overfitting - Model evaluation: MSE, RMSE, R2 score - Practical: Predict housing prices	4	16/6/2025	

Day 14	Supervised Learning - Classification	- Logistic Regression, Decision Trees - Random Forest, K-Nearest Neighbors - Confusion Matrix, ROC curve, Precision, Recall, F1-score - Application: Email spam detection	4	17/6/2025	
Module-4 Assessment Supervised Learning		**- Build a regression/classification model - Evaluation report submission**	**17/6/2025 2:30-4:30 (Group-A) 10:00-12:00 (Group-B)**		
Day 15	Unsupervised Learning - Clustering	- K-Means algorithm, elbow method - DBSCAN and Hierarchical clustering - Evaluating clustering: silhouette score, Davies–Bouldin index	4	18/6/2025	
Day 16	Dimensionality Reduction	- Introduction to high-dimensional data problems - PCA: Variance, components, implementation - t-SNE: Visualizing high-dimensional data	4	19/6/2025	
Module-5 Assessment Unsupervised Learning		**- Hands-on clustering task using real data - PCA implementation and result analysis**	**19/6/2025 2:30-4:30 (Group-A) 10:00-12:00 (Group-B)**		
Day 17	Introduction to Deep Learning	- Overview of Deep Learning vs ML - ANN Architecture: layers, neurons, weights, activation functions - Cost functions, backpropagation - TensorFlow and Keras: model creation, compiling, fitting	4	20/6/2025	
Day 18	CNNs and RNNs	- CNN: Convolution, pooling, feature maps - Image classification with CNN (e.g., MNIST) - RNN: Sequential modeling, LSTM networks - Time series prediction basics	4	21/6/2025	
Day 19	NLP	Introduction and Application of Natural Language Processing	4	23/6/2025	
Module -6 Assessment Deep Learning		**- Build and evaluate a CNN on image dataset - Submit model architecture and performance metrics**	**23/6/2025 2:30-4:30 (Group-A) 10:00-12:00 (Group-B)**		
Day 20 - 25	Capstone Project and Deployment	- Choose a dataset and define a business problem - Perform full lifecycle: EDA, modeling, evaluation - Deployment: Streamlit or Flask - GitHub submission and final presentation to panel	24	24/6/2025 to 29/6/2025	
Signature of Trainer		**SPOC Signature & Collage Seal**			